Populations

Exam code: 0610 & 0970

Download PDF

Written by: Lára Marie McIvor
Reviewed by: Lucy Kirkham
Updated on 24 September 2024

Definition of Population

- A population is defined as **a group of organisms of one species, living in the same area at the same time**

Other important definitions

- A community is defined as **all of the populations of different species in an ecosystem**
- An ecosystem is defined as **a unit containing the community of organisms and their environment, interacting together** (eg a decomposing log, a lake)

Population Growth

- All living organisms compete with each other for **food, water and living space**
- Those which are the best adapted to their environments generally increase their populations at the expense of those less well adapted
- Population growth in most organisms is controlled by the following three factors:
  - **Food supply**
  - **Predation**
  - **Disease**

The Population Growth Curve: Extended

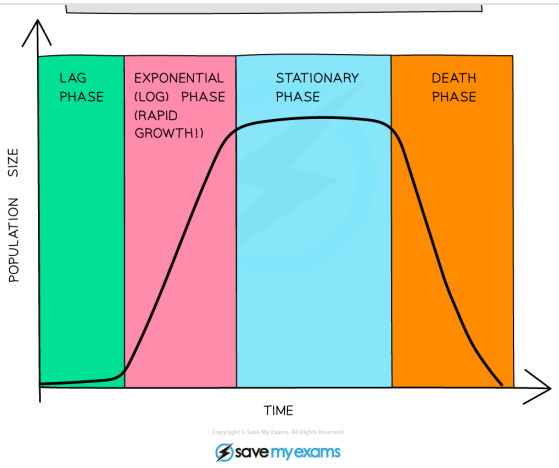
- If the growth of **microorganisms in a fermenter** is measured over time, the population growth looks like the graph below

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



A typical growth curve for a population in an enclosed environment

- The shape of this curve (a little like an 'S'), gives it its name – a **sigmoid growth curve**
- The curve has four distinct phases:
  - **Lag phase** – organisms are **adapting to the environment** before they are able to reproduce; in addition, at this stage there are very few organisms and so reproduction is not producing larger numbers of offspring
  - **Log phase** (aka **exponential** phase) – **food supply is abundant**, **birth rate is rapid** and **death rate is low**; **growth is exponential** and only limited by the number of new individuals that can be produced
  - **Stationary phase** – population **levels out** due to a **factor in the environment**, such as a nutrient, **becoming limited** as it is not being replenished; birth rate and death rate are equal and will remain so until either the nutrient is replenished or becomes severely limited
  - **Death phase** – population **decreases** as death rate is now greater than birth rate; this is usually because **food supply is short** or **metabolic wastes produced by the population have built up to toxic levels**

- Organisms in a natural environment are **unlikely to show population growth like a sigmoid growth curve** because they are affected by many other factors, including:
  - **changing temperature or light**
  - **predators**
  - **disease**
  - **immigration** (individuals moving into the area)
  - **emigration** (individuals moving out of the area)

Teacher only

New! Exclusive Teacher-Only Questions

Fresh, expert-written questions

Explore in Test Builder

Test yourself

Flashcards

Was this revision note helpful? Yes No

Previous

Next

Course

Study tools

All topics

Resources

Learning Hub

Scholarship

Join

Past Papers

Solution Banks

Sitemap

Dashboard

Overview

Brand ambassador

Profile

Subscription

Settings

Log out

Company

About us

Exam Specificity

Content Quality

Promotions

Jobs

Terms

Privacy

Cookie Policy

Help and Support

Subjects

Biology

Chemistry

Physics

Maths

Geography

English Literature

Psychology

All Subjects